PHY-112 | Principles of Physics-2

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Department of Mathematics & Natural Sciences

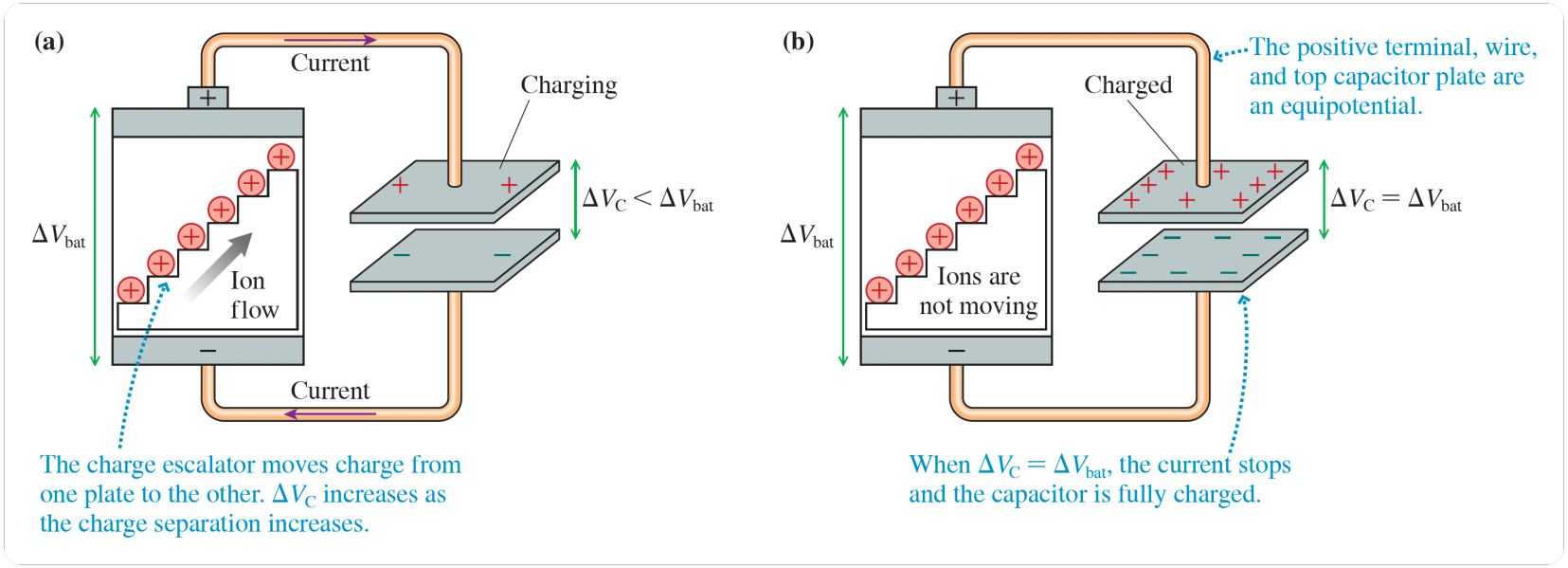
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Tour de Capacitors and Capacitance

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How to Create a Capacitor?

Using what we know so far!



What is Capacitance of a Capacitor?

It is limited by its dimensions

$$C = \frac{Q}{\Delta V_C}$$

Capacitance of a Capacitor

- measures the capacity of storing energy, not charge.
- is a scalar; measured in $\mathrm{C\,V^{-1}}$, or commonly, **Farad** [F] unit.
- typically μF or nF is used in practice.
- stores energy in the $\vec{E}$-field in between the plates.
- is restricted by the capacitor's geometry/dimensions.

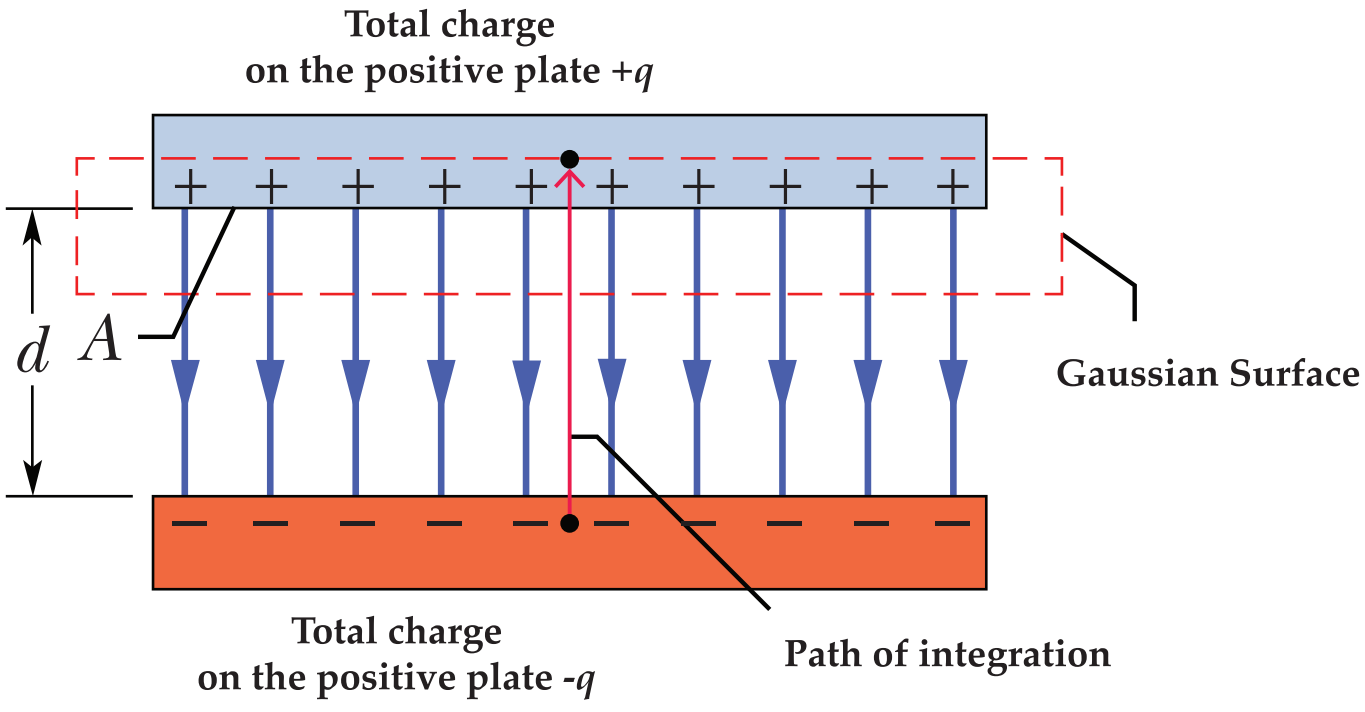
Parallel-Plate Capacitors

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What is even a Parallel-Plate Capacitor?

The plates are two planar surfaces separated by some distance

Parallel-plate Capacitor



End/Side view

Capacitance of a Parallel-Plate Capacitor

The plates are two planar surfaces separated by some distance

$$C = \frac{Q}{\Delta V_C} = \frac{\epsilon_0 EA}{\int_-^+ Edr}.$$

$$\Delta V_C = E \int_-^+ dr = Ed.$$

$$C = \frac{\epsilon_0 A}{d}.$$

Incepting Ideas (1)

Hint: Use the formal definition for capacitance

Q1: The parallel plates of a 1.0 F capacitor are 1.0 mm apart. What is their area?

Q2: The spacing between the plates of a 1.0 mF capacitor is 0.050 mm.

- What is the surface area of the plates?
- How much charge is on the plates if this capacitor is charged to 1.5 V?

Q3: The plates of a parallel-plate capacitor in vacuum are 5.00 mm apart and 2.00 m^2 in area. A 10.0 kV potential difference is applied across the capacitor. Compute

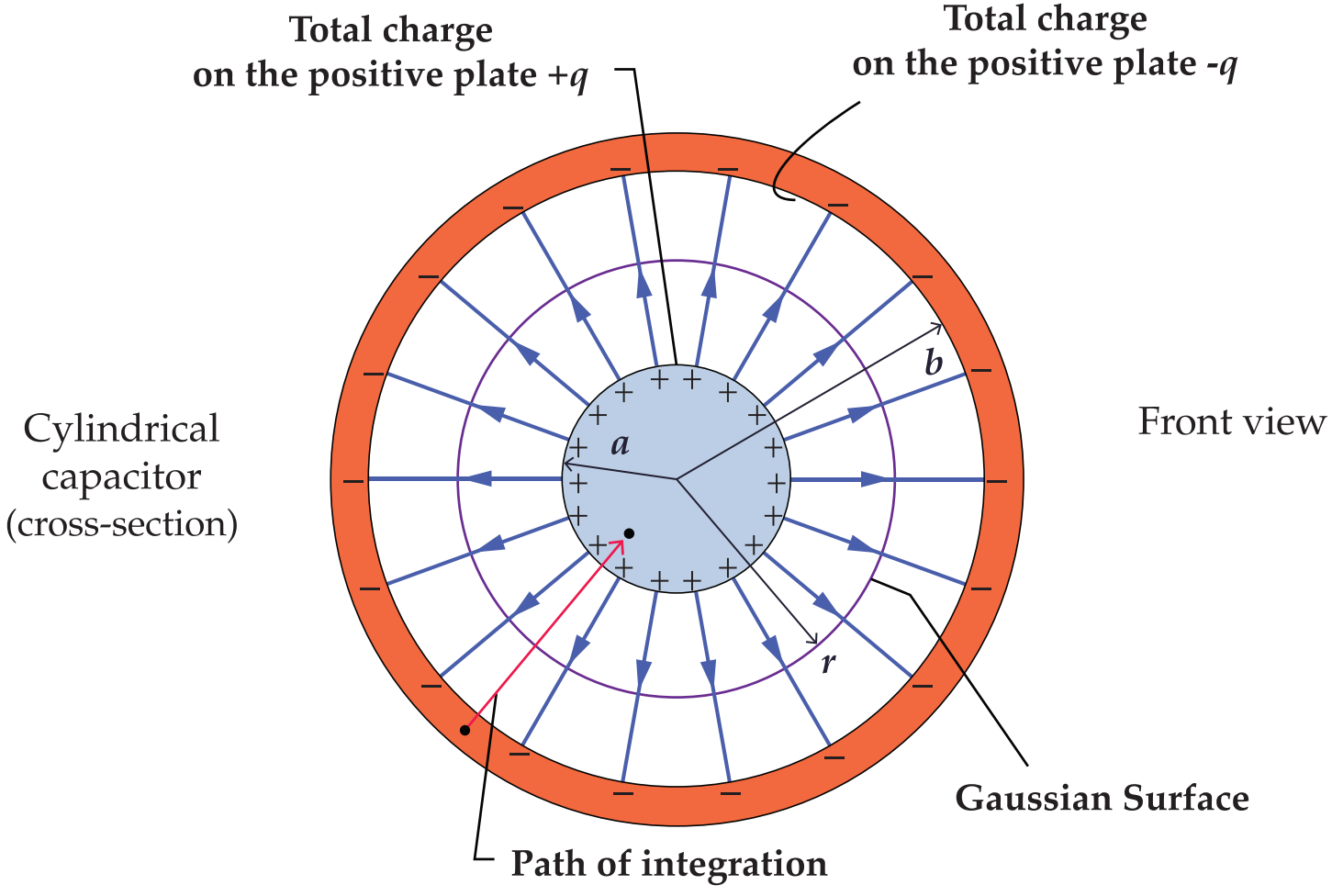
- the capacitance,
- the charge on each plate,
- the magnitude of the electric field between the plates.

Cylindrical Capacitors

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What is even a Cylindrical Capacitor?

The plates are wrapped around



Capacitance of a Cylindrical Capacitor

The plates are two coaxial cylinders

$$C = \frac{Q}{\Delta V_C} = \frac{\epsilon_0 EA}{\int_-^+ E dr}.$$

$$Q = \epsilon_0 E \cdot 2\pi r L.$$

$$\Delta V_C = -\frac{\lambda}{2\pi\epsilon_0}\int_+^- \frac{1}{r} dr = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right).$$

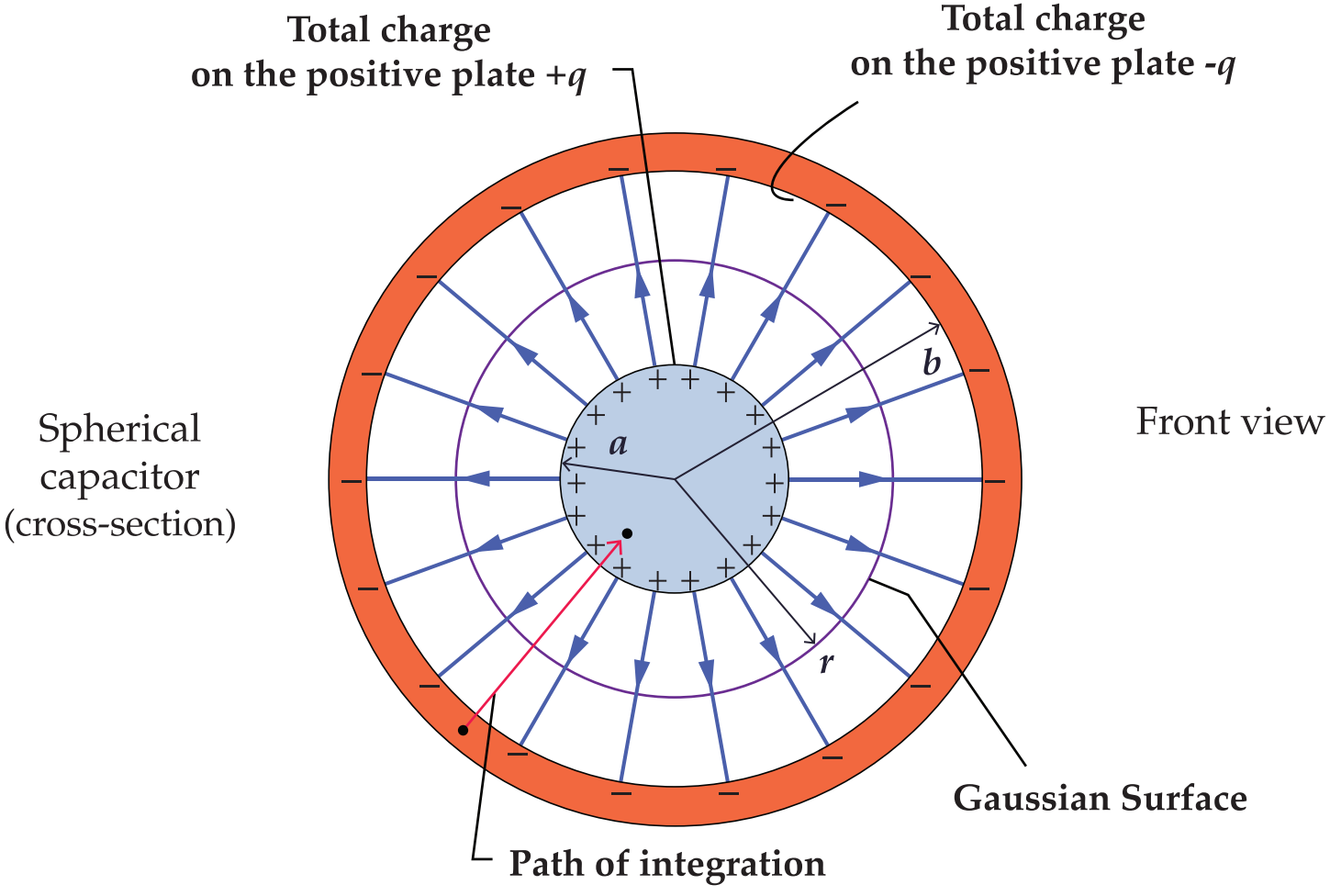
$$C = 2\pi\epsilon_0 \frac{L}{\ln\left(\frac{b}{a}\right)}.$$

Spherical Capacitors

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What is even a Spherical Capacitor?

The plates are two concentric spherical shells



Capacitance of a Spherical Capacitor

The plates are two concentric spherical shells

$$C = \frac{Q}{\Delta V_C} = \frac{\epsilon_0 EA}{\int_-^+ E dr}.$$

$$Q = \epsilon_0 E \cdot 4\pi r^2.$$

$$\Delta V_C = -\frac{Q}{4\pi\epsilon_0}\int_+^- \frac{1}{r^2}dr = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right) = \frac{Q}{4\pi\epsilon_0}\cdot\frac{b-a}{ab}.$$

$$C = 4\pi\epsilon_0\left[\frac{ab}{b-a}\right].$$

Incepting Ideas (2)

Hint: Use the formal definition for capacitance

Q1: A capacitor is made from two hollow, coaxial, iron cylinders, one inside the other. The inner cylinder is negatively charged and the outer is positively charged; the magnitude of the charge on each is 10.0 pC. The inner cylinder has radius 0.50 mm, the outer one has radius 5.00 mm, and the length of each cylinder is 18.0 cm. (a) What is the capacitance? (b) What applied potential difference is necessary to produce these charges on the cylinders?

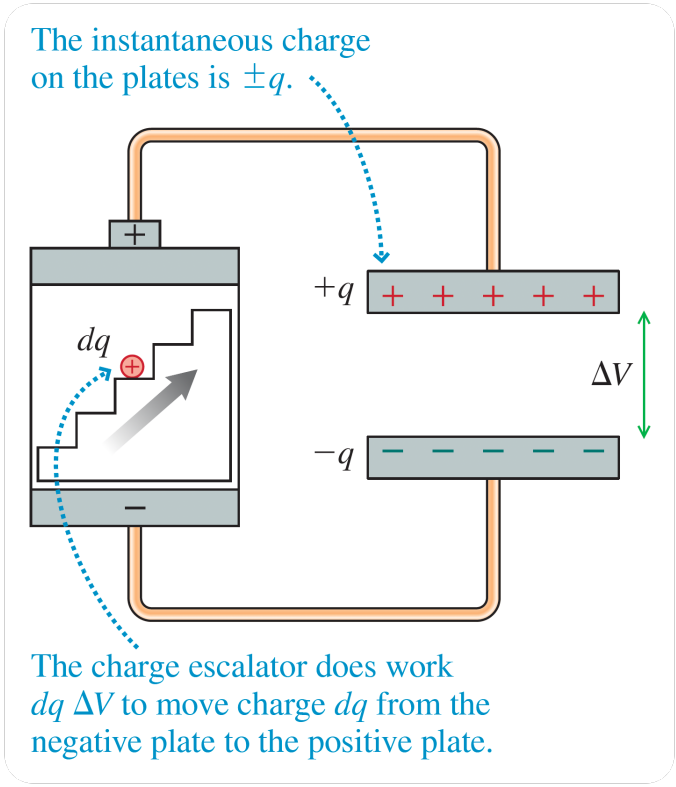
Q2: A spherical capacitor is formed from two concentric, spherical, conducting shells separated by vacuum. The inner sphere has radius 15.0 cm and the capacitance is 116 pF. (a) What is the radius of the outer sphere? (b) If the potential difference between the two spheres is 220 V, what is the magnitude of charge on each sphere

The Energy Stored in a Capacitor

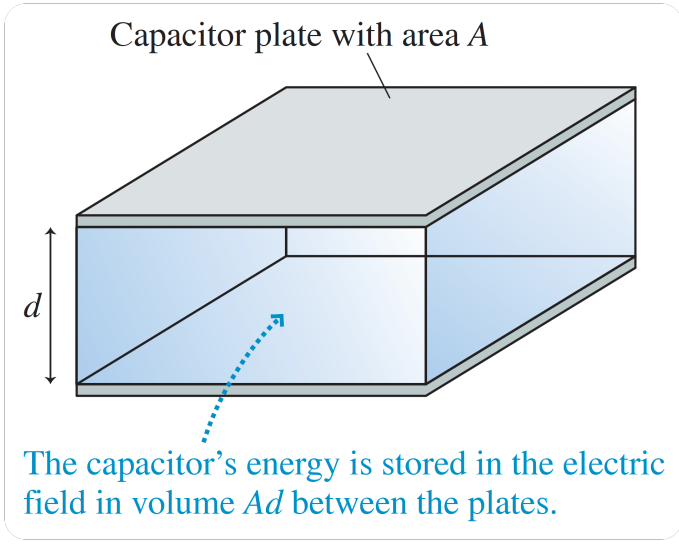
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The Energy Stored in a Capacitor

It is stored in the fields



(a)



(b)

The Energy Stored in a Capacitor

It is stored in the fields

$$dU = dq\Delta V_C = dq\frac{q}{C}$$

$$U_C = \int dU = \frac{1}{C}\int_0^Q q\,dq$$

$$U_C = \frac{Q^2}{2C}$$

$$= \frac{Q}{2} \times \frac{Q}{C} = \frac{Q}{2}\Delta V_C$$

$$= \frac{Q^2}{C^2} \times \frac{C}{2} = \frac{1}{2}C\Delta V_C^2$$

$$U_C = \frac{Q^2}{2C} = \frac{1}{2}Q\Delta V_C = \frac{1}{2}C\Delta V_C^2$$

The Energy Density in a Capacitor

Quantity looks the same for all shapes

$$\text{Energy, } U_C = \frac{1}{2}\epsilon_0 V E^2 \ \ [\text{J}]$$

$$V_{\text{parallel-plate}} = Ad \ \ [\text{m}^3]$$

$$V_{\text{cylindrical-plate}} = 2\pi(b-a)L \ \ [\text{m}^3]$$

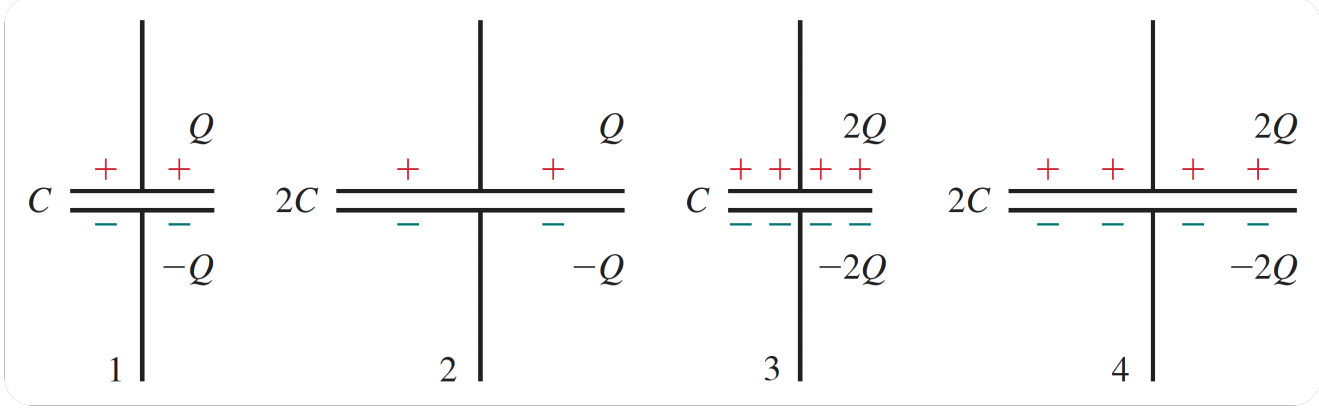
$$V_{\text{spherical-plate}} = \frac{4\pi}{3}\left(b^3 - a^3\right) \ \ [\text{m}^3]$$

$$\text{Energy Density, } u_C = \frac{U_C}{V} = \frac{1}{2}\epsilon_0 E^2 \ \ [\text{J m}^{-3}]$$

Incepting Ideas (3)

Hint: Use either of the three energy formula

Q1: Of the four capacitors (All four are kept at d separation), rank in order, from largest to smallest,



- the potential differences $(\Delta V_C)_1$ to $(\Delta V_C)_4$
- the maximum storable energies U_1 to U_4.

Q2: How much energy is stored in a 220 mF camera-flash capacitor that has been charged to 330 V?

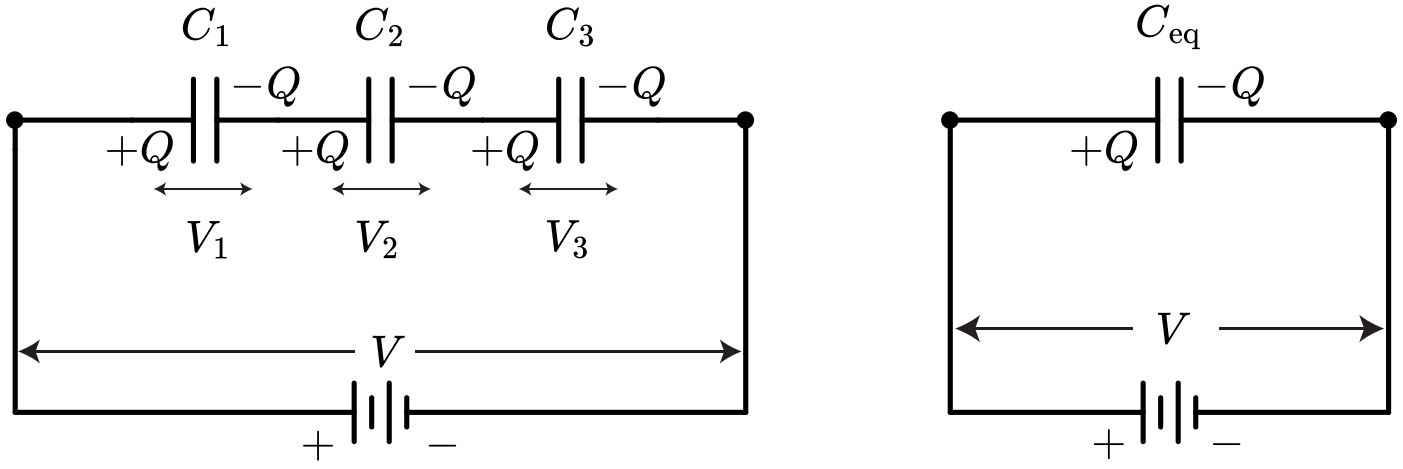
Capacitors in Series and Parallel

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Combinations of Capacitors: Series

Try this at home

The voltage across each capacitor is not the same. The total voltage across the series combination equals the sum of the voltages across each capacitor.

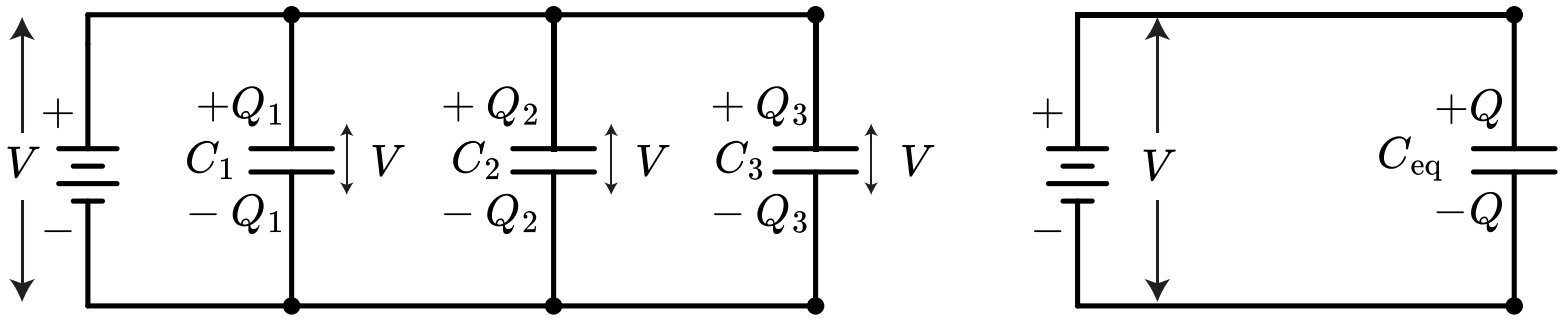


$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} = \sum_i^n \frac{1}{C_i}$$

Combinations of Capacitors: Parallel

Try this at home

The voltage across each capacitor is the same. The total charge stored in the combination is the sum of the charges stored in each individual capacitor.

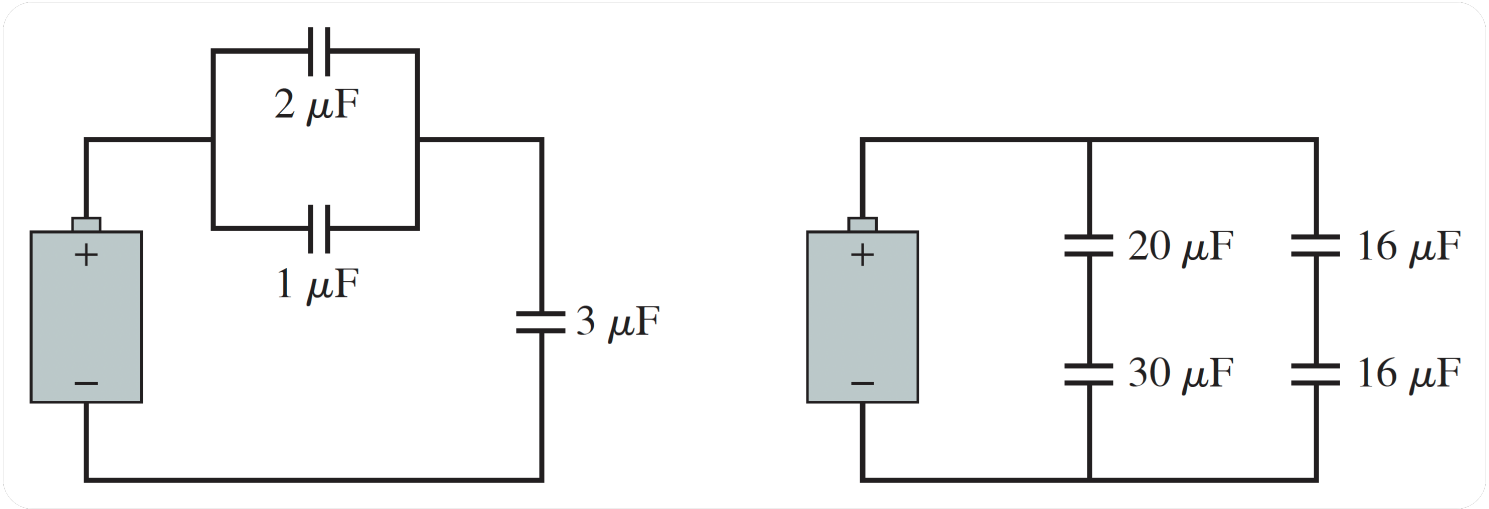


$$C_{eq} = C_1 + C_2 + C_3 = \sum_{i}^{n} C_i$$

Incepting Ideas (4)

Watch out for nodes

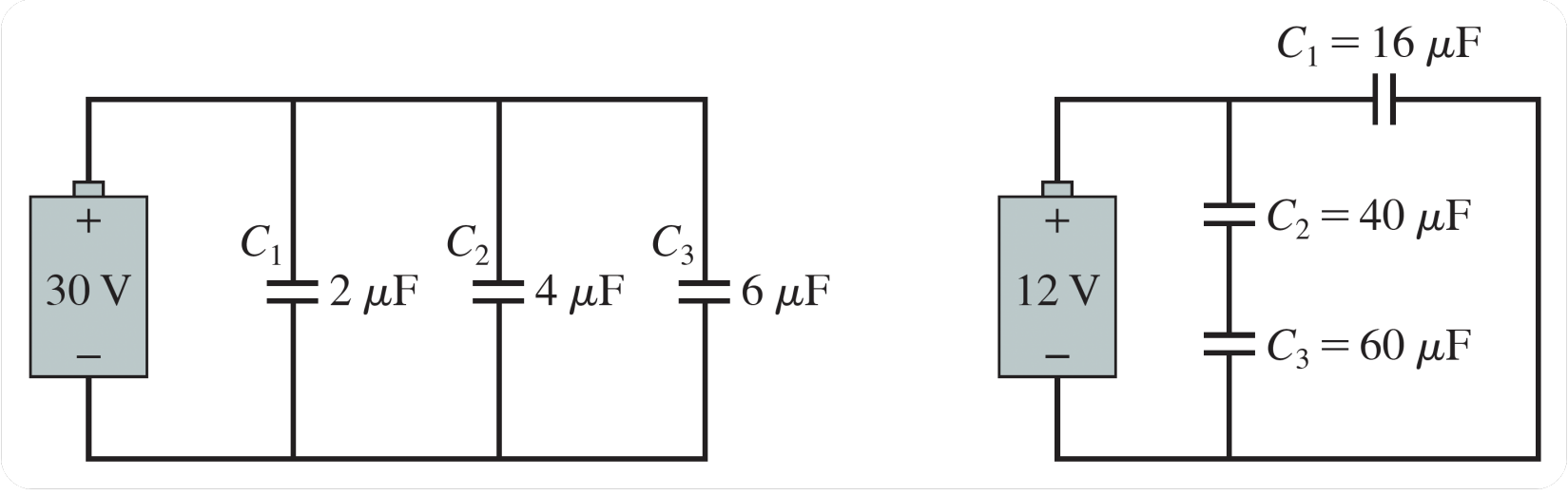
Q: What is the equivalent capacitance of the following two circuits?



Incepting Ideas (5)

Simplify first and then circle back

Q: What are the charges and potential differences across each capacitor?



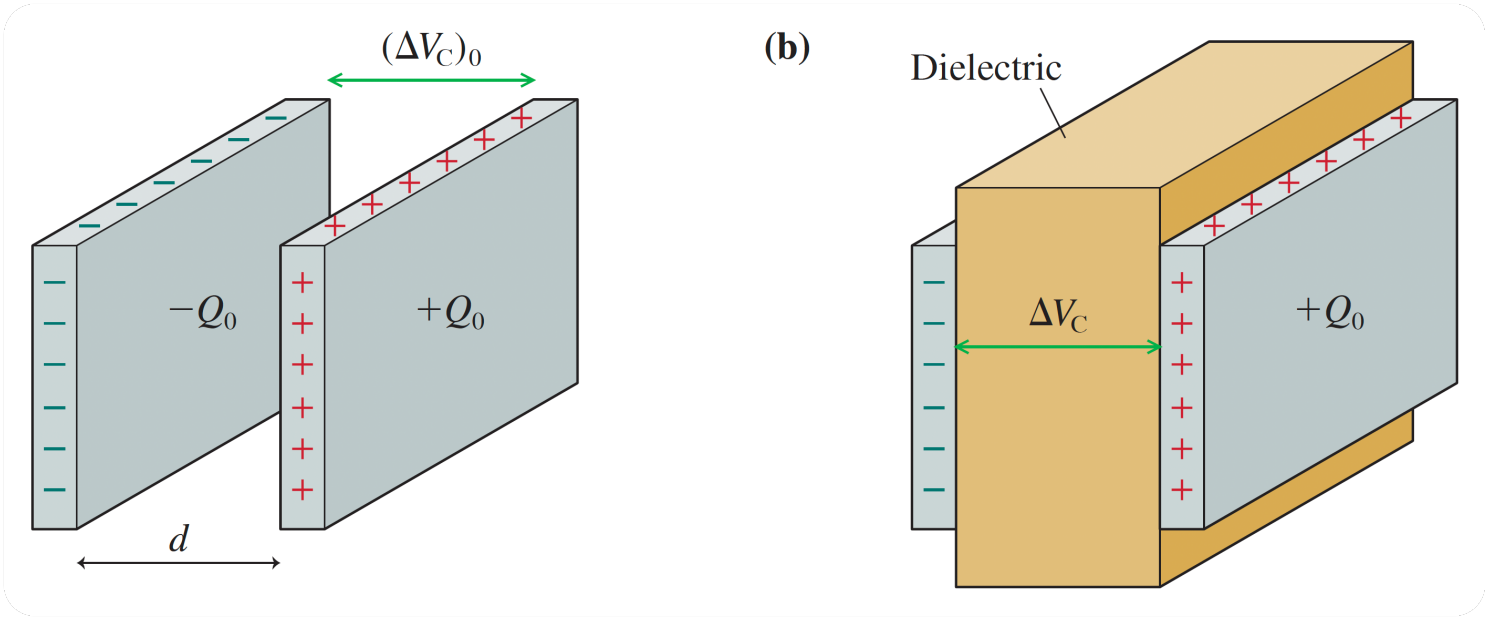
Capacitors with Dielectric Materials

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Dielectrics in Capacitor

Dielectrics are insulating materials

Vacuum-insulated and Dielectric filled Capacitors



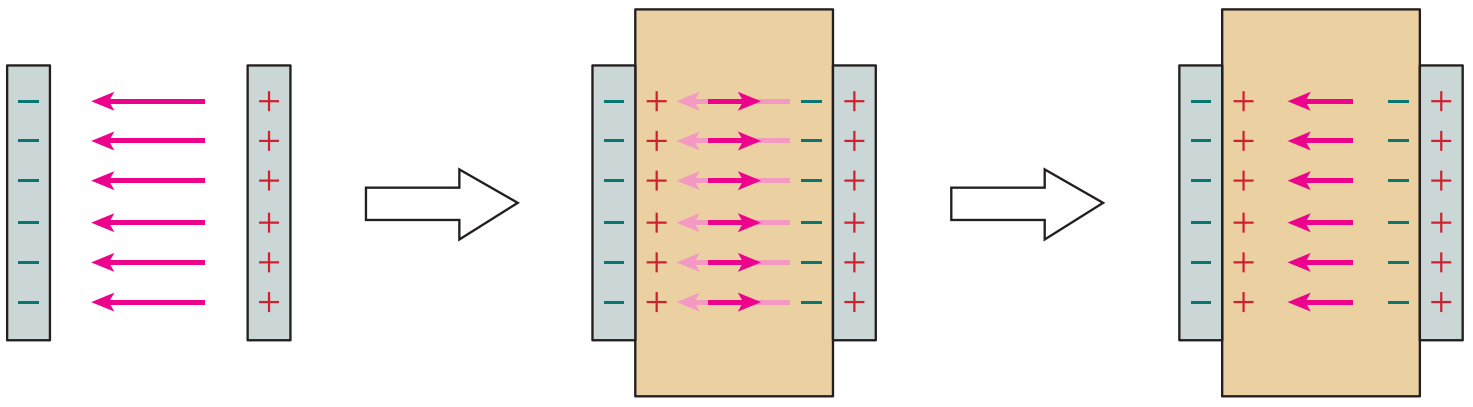
Consequences of filling a Capacitor with a Dielectric

The internal dipoles create an opposing electric field

$$E_0 = \frac{\sigma_0}{\epsilon_0}$$

$$E_{\text{ind}} = \frac{\sigma_{\text{ind}}}{\epsilon_0}$$

$$E_{\text{new}} = E_0 - E_{\text{ind}}$$



Consequences of filling a Capacitor with a Dielectric

Vacuum is the baseline dielectric

Define the Dielectric Constant

$$\kappa = \frac{E_0}{E_{\text{new}}} = \frac{(\Delta V_C)_0}{\Delta V_C)_{\text{new}}} = \frac{C_{\text{new}}}{C_0}$$

The Permittivity of Free Space

A fundamental constant with physical significance

$$\kappa = \frac{C_{new}}{C_0} = \frac{\frac{\epsilon A}{d}}{\frac{\epsilon_0 A}{d}}$$

$$\kappa = \frac{\epsilon}{\epsilon_0} = \frac{E_0}{E_{new}}$$

$$\epsilon = \kappa \epsilon_0$$

A New Set of Capacitor Equations

With dielectric materials squeezed inside

Capacitance of parallel-plate capacitors, $C = \frac{\epsilon A}{d} = \frac{(\kappa\epsilon_0)A}{d}$.

Capacitance of cylindrical capacitors, $C = 2\pi\epsilon\frac{L}{\ln\left(\frac{b}{a}\right)} = 2\pi(\kappa\epsilon_0)\frac{L}{\ln\left(\frac{b}{a}\right)}$.

Capacitance of spherical capacitors, $C = 4\pi\epsilon\left[\frac{ab}{b-a}\right] = 4\pi(\kappa\epsilon_0)\left[\frac{ab}{b-a}\right]$

Energy Density, $u_C = \frac{1}{2}\epsilon E^2 = \frac{1}{2}(\kappa\epsilon_0)E^2$.

Incepting Ideas (6)

Hint: Dielectric placed inside a parallel-plate capacitors

Q: A 5.0 nF parallel-plate capacitor is charged to 160 V. It is then disconnected from the battery and immersed in distilled water ($\kappa = 78.57$). What are—

- the capacitance and voltage of the water-filled capacitor
- the energy stored in the capacitor before and after its immersion?

Incepting Ideas (7)

Hint: Dielectric placed inside a parallel-plate capacitors

Q: A chest defibrillator unit contains a 150 mF capacitor that is charged to 2100 V. The capacitor plates are separated by a 0.050 mm-thick insulator with dielectric constant 120.

- What is the area of the capacitor plates?
- What are the stored energy and the energy density in the electric field when the capacitor is charged?

Incepting Ideas (8)

Hint: Practice! Practice! Practice!

Some Problems to Practice at Home on Capacitance

Q-24.7, 24.8 p-833, **Example** 24.1, 24.2, p-812 | Young-Freedman
Exercise Problem 26.1, 26.2, 26.3, p-833 | YF
Checkpoint 1, p-775, 2, p-777, 3, p-779 **Sample Problem** 25.03, p-727, 25.04, p-730 | Resnick-Halliday
Exercise Problem 6, p-739, 12, p-740 | RH

That is it for today!

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